

QATIPv3 – Azure Lab5a

Migrating a local Statefile to an Azure Storage Account

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Lab Objectives

In this lab you will:

1. Deploy resources using the provided code, creating a statefile on your local device.
2. Reconfigure the code to migrate the statefile to a central Azure storage account.

Teaching Points

This lab will use terraform to deploy a network stack into Azure. This will create a local backend (Statefile) on the client device used for the deployment. This Statefile will then be migrated to a centrally controlled secure backend, an Azure storage account, as is more typical in a production environment.

Before you begin

1. Ensure you have completed Lab0 before attempting this lab.
2. In the IDE terminal pane, enter the following command...

cd c:\azurelabs\lab5

3. This shifts your current working directory to lab5. Ensure all commands are executed in this directory
4. Close any open files and use the Explorer pane to navigate to and open the lab5 folder.
5. Update line **18** in **main.tf** with your lab subscription id and save the change.
6. Run **terraform init** and then **terraform plan** commands
7. Verify there are no errors and then run **terraform apply** followed by **yes**
8. Switch the console to review the deployment

You can carry on with the lab whilst the deployment progresses as this may take several minutes. You are deploying the following resources into the East US region; a resource group “RG5” containing a VNet with 2 subnets, a Public IP address and 2 virtual machines. A total of 12 resources will be created. Note that a URL will be displayed as output. This is not functional at the moment and will be used in lab5b.

Solution

The solution to this lab can be found in c:\azurelabs\solutions\lab5. Try to use this only as a last resort if you are struggling to complete the step-by-step processes.

Task1: Create a Storage Account for Remote State

In this task you will create an Azure storage account and blob container for use as a terraform remote backend.

1. Navigate to the Azure Portal.
2. Search for and select **Storage Accounts**.
3. Click **Create** to create a new storage account....

Resource Group: create new **RG1**

Storage Account Name: **remotestate<your-name>** , replacing `**<your-name>**` with a unique identifier.

Region: **East US**

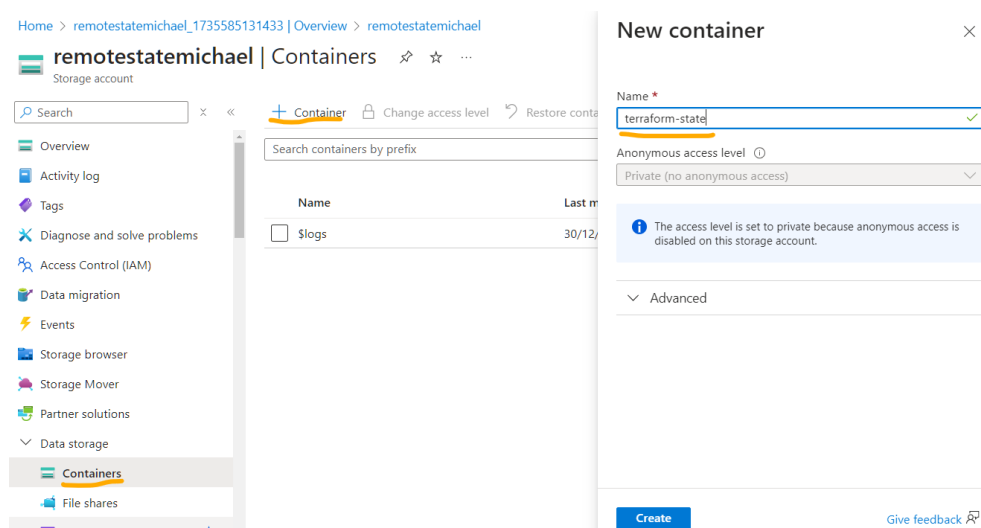
Performance: Standard

Replication: Locally-redundant storage (LRS)

4. Leaving other settings as defaults, click **Review + Create** and then **Create**.
5. Once the storage account is created, navigate to it and create a new **Blob Container**:

Name: terraform-state

Public access level: Private

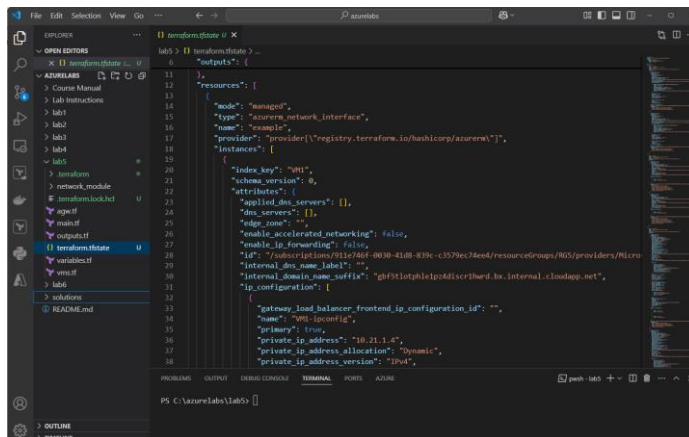


Task2: Review the Local Terraform State

1. Wait until your deployment has completed..

Apply complete! Resources: 12 added, 0 changed, 0 destroyed.

2. Using VSC Explorer, select and review the newly created terraform.tfstate file in the lab5 folder...



Unless otherwise specified, the state file for your deployments will be in the local root directory. In all but trivial deployments the state file will be stored in a remote, shared location. You will move yours to the storage account you created in Task 1.

Task3: Update the Terraform Configuration

1. Open main.tf in the root lab5 folder.
2. Add the following backend configuration for Azure code to the end of the file, replacing <your storage account> with the name of the storage account you created in Task 2 above ...

```
terraform {
  backend "azurerm" {
    resource_group_name = "RG1"
    storage_account_name = "<your storage account name>"
    container_name = "terraform-state"
    key = "terraform.tfstate"
  }
}
```

3. Save the file.

Task4: Initialize the Azure Backend and verify migration

1. Run the following command to migrate the state: **terraform init**

2. During the initialization, Terraform will prompt to confirm the migration of the local state to Azure. Type **yes** when prompted.
3. Confirm that the state has been migrated by checking the blob container for the terraform.tfstate file. Make a note of its size. In Lab5b you will deploy additional resources, and this will be reflected by a growth in the state file size. Also notice that your local state file is now empty.

Name	Modified	Access tier	Archiv...	Blob type	Size
☐ terraform.tfstate	27/12/2024, 21:46:25	Hot (Inferred)		Block blob	24.13 KiB

4. Do not delete any resources as they will be used in the next lab

***** Congratulations, you have completed this lab *****